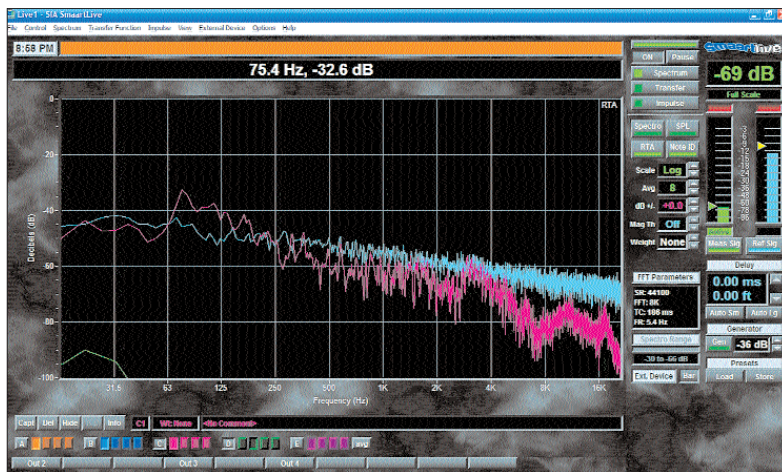




The Artis S6600R/FM fails a little at reproduction of higher pitched sounds—between the 4 and 16 KHz mark

A peculiar thing we noticed was that the Tech-Com SSD-5101R and the Altec Lansing VS3251 could not reproduce piano sounds in 2.1 mode. When we switched to 5.1 mode, only the centre speaker reproduced the piano, but the bass was lost.

DVD Movies

The Artis X10 Mini and the S6600R/FM and S8800 were very good in the DVD movie test. The bass, treble, and mid-range were all just about perfect. But once we plugged in the Zebronic ZEB-SW13100R, we found that the set possessed the power to edge past the former speaker sets,

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Artis S6600R/FM
Everything you ever needed—it's all here

and we had to raise the bar we had set for the benchmarks. The explosions seemed earth-shatteringly real—we could feel the shrapnel flying around us! The engine roar was surreal, and the concussion gun almost gave us a concussion!

Other brands such as Altec Lansing and Tech-Com were decent, but the Altec Lansing FX5051 produced overwhelmingly overpowering bass, which eclipsed some sound detail.

DVD Audio

The Eric Clapton song was rendered beautifully on the Zebronic ZEB-SW13100R and the Artis S6600R/FM and X10 Mini. The ZEB-SW13100R and X10 Mini produced richer bass, while the S6600R/FM produced crisper highs. The sets from Altec Lansing and Tech-Com also gave decent performance and are worthy mentions.

Game Performance

The Artis S6600R/FM, X10 Mini, and the S8800 were very good in the game test and could create an almost-real virtual environment around us. But the Zebronic ZEB-SW13100R was in a

Size Does Matter

The normal human ear can perceive sound from 20 Hz to 20 KHz. Ideally, a single speaker should be able to reproduce this entire range of the aural spectrum. But as is evident, in real-life scenarios, this is not possible. To maintain the output level, for every halving of a given frequency, the speaker must double its excursion. For the layman, this means that if a speaker diaphragm moves 1 mm at 20 KHz (20,000 Hz), it would have to move a distance of 1 km to be able to produce the same level of sound at 20 Hz. Unrealistic, isn't it?

To overcome this problem, we have speakers built to operate in different frequency ranges. Depending on need, this is divided into three or four frequency ranges. The commonly-used frequency ranges and their significance are as below.

2,500 Hz to 20,000 Hz: This range is covered by a class of drivers called tweeters. These are typically dome-shaped and are generally 1 inch in diameter. The wavelength of a tweeter is also very small, typically 0.7 inch for 20,000 Hz. Harsh, high-pitched and ear-piercing sounds, such as a cricket chirping, electric guitars, etc., constitute this frequency range.

400 Hz to 2,500 Hz: This range is covered by mid-range speakers. Such speakers are cone-type, having a separate cone body and surrounding material. They are sized from 3 to 8 inches in diameter. The most common diameter is around 5.25 inches. This is a frequency range to which the human ear is better tuned and thus, more receptive. Therefore, most musical instruments create sounds that fall in this range.

100 Hz to 400 Hz: Mid-bass drivers cover this range. These are very much similar to mid-range speakers in features and construction, but the difference lies in the size, these being larger and designed for higher excursion. The typical size of these speakers ranges from 6.5 to 10 inches in diameter, while the most common size is 6.5 inches. Some systems incorporate speakers with diameters ranging from 5.25 to 6.5 inches and in such cases, a single driver can cover a frequency range of 100 Hz to 2,500 Hz. Male voices, and instruments such as the cello, bassoon, and French horn are some examples for this frequency range.

20 Hz to 100 Hz: This is the lowest audible frequency range covered by speakers. Known as bass, the speakers designed to cover this range are known as woofers. Sharing the same features and construction as mid-range and mid-bass class of speakers, woofers have the lowest of the resonant frequencies and the highest of the excursion capacities of the three. The typical size of woofers ranges from 6 inches to as large as 36 inches. The average size of woofers is 10 to 12 inches. Kick drums, organs, contrabassoon and string bass are some of the sources of sounds in this range.

A typical speaker system consists of at least two of the above types of speakers, tweeters and woofers being those types. The woofer usually operates in the 20 Hz to 150 Hz range, while tweeters operate in the range of 150 Hz to 20 KHz. The sound signals from the sound card are received by a special crossover circuit containing electronic filters that separate the different frequency ranges and feeds the adequate frequency range according to speaker type.

We would like to add that speaker systems that use a single subwoofer and a set of satellites can never provide you pure audiophile-quality sound. To cite one example, we would like to bring to your notice that in such a speaker system, only the subwoofer produces bass. So what about multi-channel, low-frequency sounds? That is entirely neglected. So in the case of a bass drum beating on one side of the listener or moving around, the bass portion of the sound will only be produced by the subwoofer (from one direction i.e. from the centre of course), while the satellites will produce the higher frequency sounds of the sticks striking the drum's surface from the correct direction. This will no doubt produce a pseudo-multichannel impression of the drum beats, but it definitely will not be anything near the real thing. A solution to this problem is to have the different types of speakers mentioned above, in each of the satellites, and to disband the subwoofer as a whole. This is the principle deployed by manufacturers of audiophile quality speaker sets, and is one of the reasons why such speakers are so expensive.